

Interreg



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RESPOND

Alpine Space

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EbA strategies Landscape elements



About this EbA strategy:

Landscape elements such as hedges, isolated trees, and shrubs are forms of plant diversification integrated into agricultural landscapes to enhance their ecological functions and promote sustainability. As nature-based solutions, these elements leverage ecosystem processes to provide environmental, social, and economic benefits.

Woody plants increase the compositional and configurational diversity of the cultural landscape, strengthening habitat connectivity and supporting biodiversity. They provide shelter, food resources, and breeding sites for a wide range of species, including pollinators and natural enemies of pests.

The establishment, maintenance, or conservation of landscape elements plays a key role in creating valuable ecological niches and enhancing the resilience of agricultural landscapes to environmental pressures and climate variability.

Key ecosystem services provided:

- Habitat provision and biodiversity conservation (*refuge, food resources, and ecological connectivity*)
- Microclimate regulation (*wind buffering, shading, and temperature moderation*)
- Water regulation (*improved infiltration, retention, and reduced nutrient leaching*)
- Soil health improvement (*enhanced organic matter, nutrient cycling, and soil structure*)
- Erosion control (*root stabilisation*)
- Water retention (*reduced surface runoff*)
- Natural pest control (*support for beneficial insects and natural enemies*)
- Cultural and landscape aesthetic value (*cultural identity, structural diversity and visual character of agricultural landscapes*)

Potential challenges:

- **Water competition:** While deep-rooted trees can enhance water availability through hydraulic lift, competition for water can still occur if not carefully managed.
- **Management and labour costs:** Hedges require regular and professional maintenance to ensure their long-term ecological functions.